

ID NUMBER: 202312065NAME: Shah Jinaash**SET B**NOTES:1. Total MARKS: $6 \times 3 = 18$.TIME allowed: 1 HOUR.2. Write your answers in the boxes provided below.

	ANSWER
Q-1	(a) $\Theta(V^2E)$ (b) $\Theta(V \cdot E)$
Q-2	adbeef
Q-3	weight of the minimum spanning tree = 17
Q-4	DFS (G) DFS (G, T)
Q-5	(a) 2 (b) 4
Q-6	(a) forward edge = (c, f) (b) back edge = (c, b) (c) cross edge = (d, e)

Q-1. State the running times of (a) single-source shortest path algorithm for DAGs, and (b) Bellman-Ford single-source shortest path algorithm.

Q-2. In the directed graph of Fig. 2, cross out the vertex g and the two edges (f, g) and (g, h). List the vertices of the resulting graph (with 6 vertices) in a topologically sorted order.

Q-3. For the weighted, undirected graph shown in Fig. 1, find the weight of the minimum weight spanning tree.

Q-4. In the following algorithm, two statements have been replaced by repeated letter X, with the correct indentation. Which are the two statements which appear in the algorithm in place of the two lines containing repeated letter X?

SCC (G)

1. XXXXXX;
2. Compute G^T ;
3. XXXXXX -- SUCH THAT, IN THE MAIN LOOP, VERTICES OF G ARE CONSIDERED IN DECREASING ORDER OF $f[u]$ found in step 1;
4. RETURN (TREES OBTAINED IN STEP 3);

Q-5. Perform BFS on the directed graph of Fig. 2, with root node 'a'. Vertices $ADJ[\bullet]$ are to be processed in alphabetical order. Let T be the resulting BFS tree. In T, find (a) the longest path length from root 'a' to a leaf node, and (b) the number of leaf nodes.

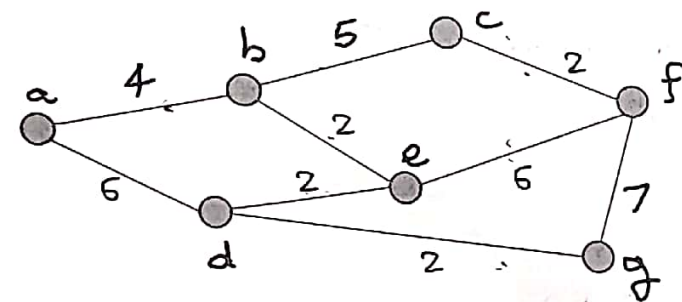


FIGURE 1

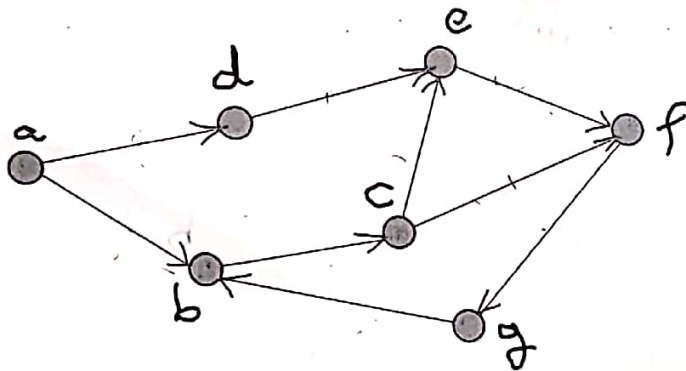


FIGURE 2

DAIICT

IT629 INTRODUCTION TO ALGORITHMS
SECOND IN-SEMESTER EXAMINATION

ID NUMBER: <u>202312062</u>	SET A
NAME: <u>Nakshi Shah</u>	

NOTES:

1. Total MARKS: 6 x 3 = 18. TIME allowed: 1 HOUR.
2. Write your answers in the boxes provided below.

	ANSWER
Q-1	$a-d=6$ $f-e=6$ $c-f=2$ $b-e=2$
Q-2	(a) 2 3 (b) 3 4
Q-3	(a) O(V+E) (a) [a,d], [b,e] (b) [f,g], [e,f], [d,e] (c) [a,b], [a,d], [b,e], [c,f], [b,c]
Q-4	(a) $O(V+E)$ (b) $O(V+E)$
Q-5	$d[v] = d[u] + 1$ $d[v] = d[u] + 1$ $d[v]$ $d[u]$ enqueue [u, w(u,v)]
Q-6	e-f-g-d-e

0
3
1
1.5
0
0

Q-1. For the weighted, undirected graph shown in Fig. 1, find the weight of the minimum weight spanning tree.

Q-2. Perform BFS on the directed graph of Fig. 2, with root node 'a'. Vertices $ADJ[\bullet]$ are to be processed in alphabetical order. Let T be the resulting BFS tree. In T, find (a) the number of leaf nodes, and (b) the longest path length from root 'a' to a leaf node.

Q-3. Perform DFS on the directed graph of Fig. 2, with root node 'a'. Vertices $ADJ[\bullet]$ are to be processed in alphabetical order. Thereby find (a) the forward edges, (b) the backward edges, and (c) the cross edges.

Q-4. State the running times of (a) Bellman-Ford single-source shortest path algorithm, and (b) single-source shortest path algorithm for DAGs.

Q-5. In the following function, two statements have been replaced by repeated letter X, with the correct indentation. Which are the two statements which appear in the function in place of the two lines containing repeated letter X?

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1. RELAX(u,v,w)
2.   if d[v] > d[u] + w(u,v)
3.     XXXXXXXXXXXXXXXXXXXX;
4.     XXXXXXXXXXXX;

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Q-6. In the directed graph of Fig. 2, cross out the vertex g and the two edges (f,g) and (g,d) . List the vertices of the resulting graph (with 6 vertices) in a topologically sorted order.

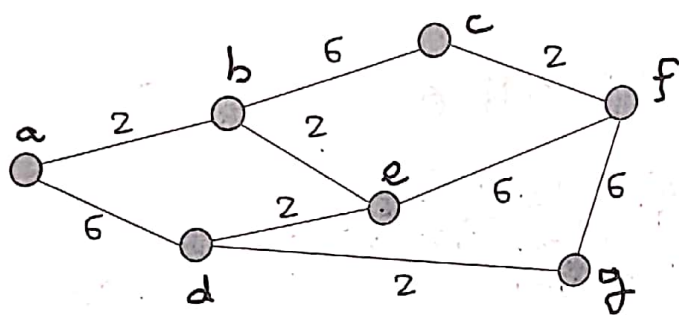


FIGURE 1

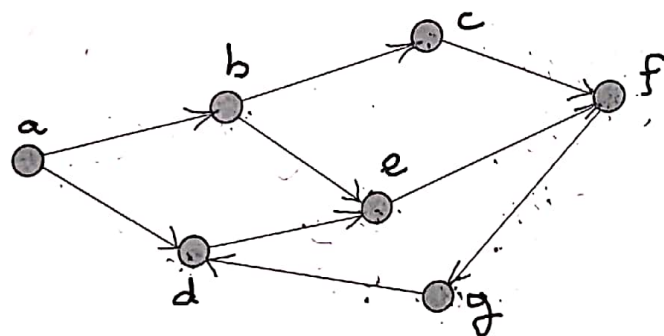


FIGURE 2